IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 28

Build a Custom VPC with Public Subnets and Internet Gateway Routing

Amazon VPC | CIDR | Public Subnets | Internet Gateway | Routing

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Portfolio documentation standard v1.0.1

Executive Summary

Built a custom **Amazon VPC** named **CloudSLAW** using the IPv4 CIDR block **10.0.0.0/16**, created two **/24 public subnets**, created and attached an **Internet Gateway** named **slaw-ig**, and updated the route table so Internet-bound traffic routes through the gateway. This replaced default-VPC exploration with a deliberately built cloud network foundation.

TERMINOLOGY	**The VPC itself uses a /16 CIDR block. The two subnets use /24 CIDR blocks inside that VPC: 10.0.1.0/24 and 10.0.2.0/24.**

Business Problem

Default VPCs are useful for quick learning, but they do not demonstrate deliberate network design. A secure AWS environment needs explicit CIDR planning, subnet placement, routing decisions, and gateway attachment. This project creates the first custom VPC baseline that future labs can extend into private subnets, NAT routing, flow logs, and workload segmentation.

Objectives

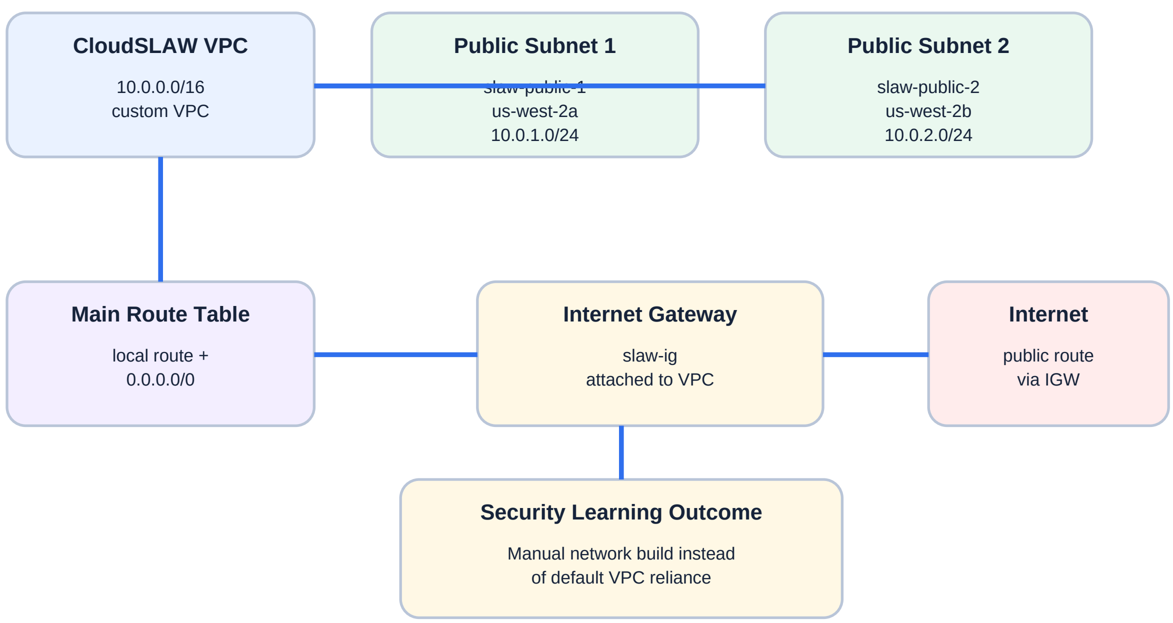
- Create a custom VPC rather than relying on the default VPC.
- Use a /16 IPv4 CIDR block for the VPC address space.
- Create two /24 public subnets inside the custom VPC.
- Place the subnets in separate Availability Zones for basic regional distribution.
- Create an Internet Gateway named slaw-ig.
- Attach the Internet Gateway to the custom VPC.
- Edit the route table to add a 0.0.0.0/0 route that targets slaw-ig.
- Confirm the completed resource map shows the VPC, subnets, route table, and network connection wired together.

Environment

Component	Configuration
Cloud Provider	Amazon Web Services (AWS)
Primary Service	Amazon VPC
Account Context	TestAccount1 / workload account
Region	us-west-2 / Oregon
VPC Name	CloudSLAW

Component	Configuration
VPC IPv4 CIDR	10.0.0.0/16
Subnet 1	slaw-public-1 - us-west-2a - 10.0.1.0/24
Subnet 2	slaw-public-2 - us-west-2b - 10.0.2.0/24
Internet Gateway	slaw-ig
Route Table	Main route table associated with both public subnets
Internet Route	0.0.0.0/0 to slaw-ig
Purpose	Bare-bones public VPC built manually for networking fundamentals

Architecture



The custom VPC uses two /24 public subnets and an Internet Gateway route for Internet reachability.

Figure 1 - Custom VPC with two public subnets, route table, and Internet Gateway.

Implementation Summary

1. Signed into the workload account with AdministratorAccess and opened VPC in us-west-2.

2. Selected VPC only so the base network could be built manually instead of using the automated VPC wizard.

3. Named the VPC CloudSLAW and set the IPv4 CIDR block to 10.0.0.0/16.

4. Created slaw-public-1 in us-west-2a with subnet CIDR 10.0.1.0/24.

5. Created slaw-public-2 in us-west-2b with subnet CIDR 10.0.2.0/24.

6. Created an Internet Gateway named slaw-ig.

7. Attached slaw-ig to the CloudSLAW VPC.

8. Edited the VPC route table and added 0.0.0.0/0 with target slaw-ig.

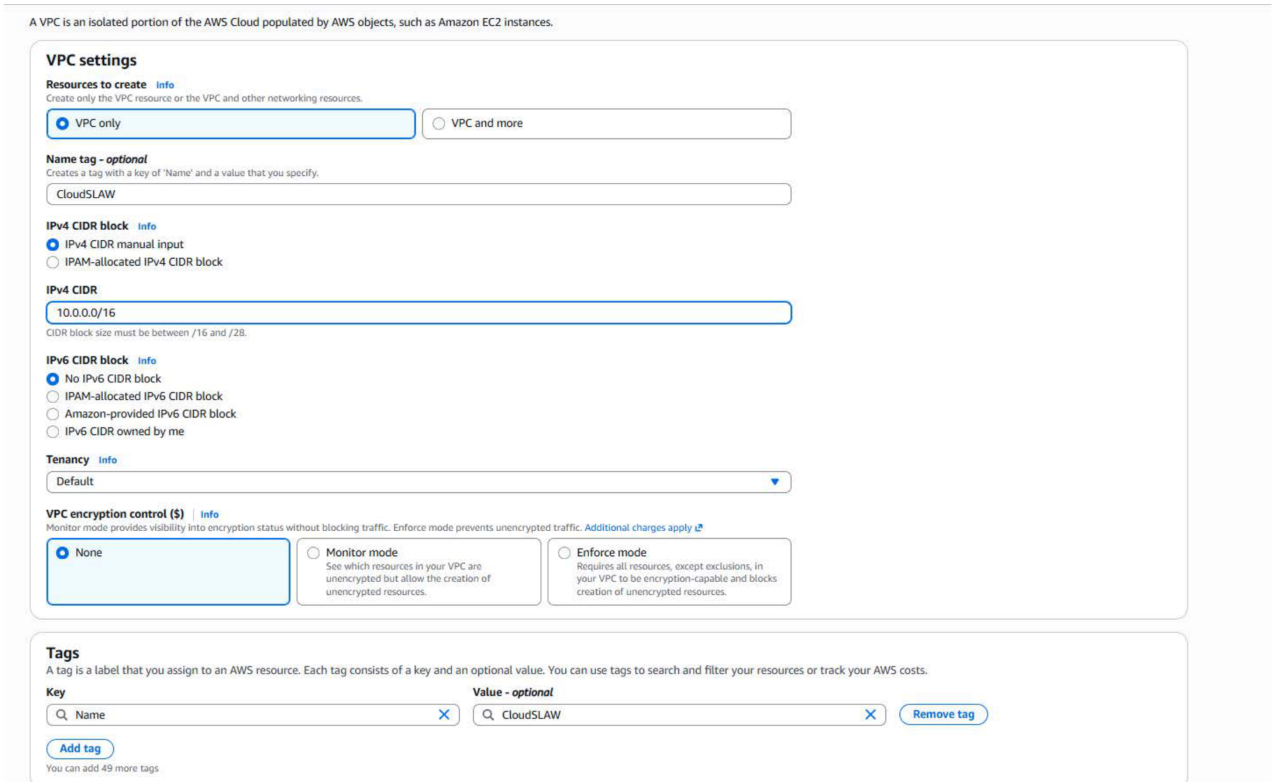
9. Reviewed the completed VPC resource map showing two public subnet associations and Internet routing through slaw-ig.

Routing Model

Route	Target	Purpose
10.0.0.0/16	local	Allows resources inside the VPC address range to route within the VPC.
0.0.0.0/0	slaw-ig Internet Gateway	Sends Internet-bound IPv4 traffic from associated public subnets to the Internet Gateway.
Subnet associations	slaw-public-1 and slaw-public-2	Makes both subnets use the public route table behavior.

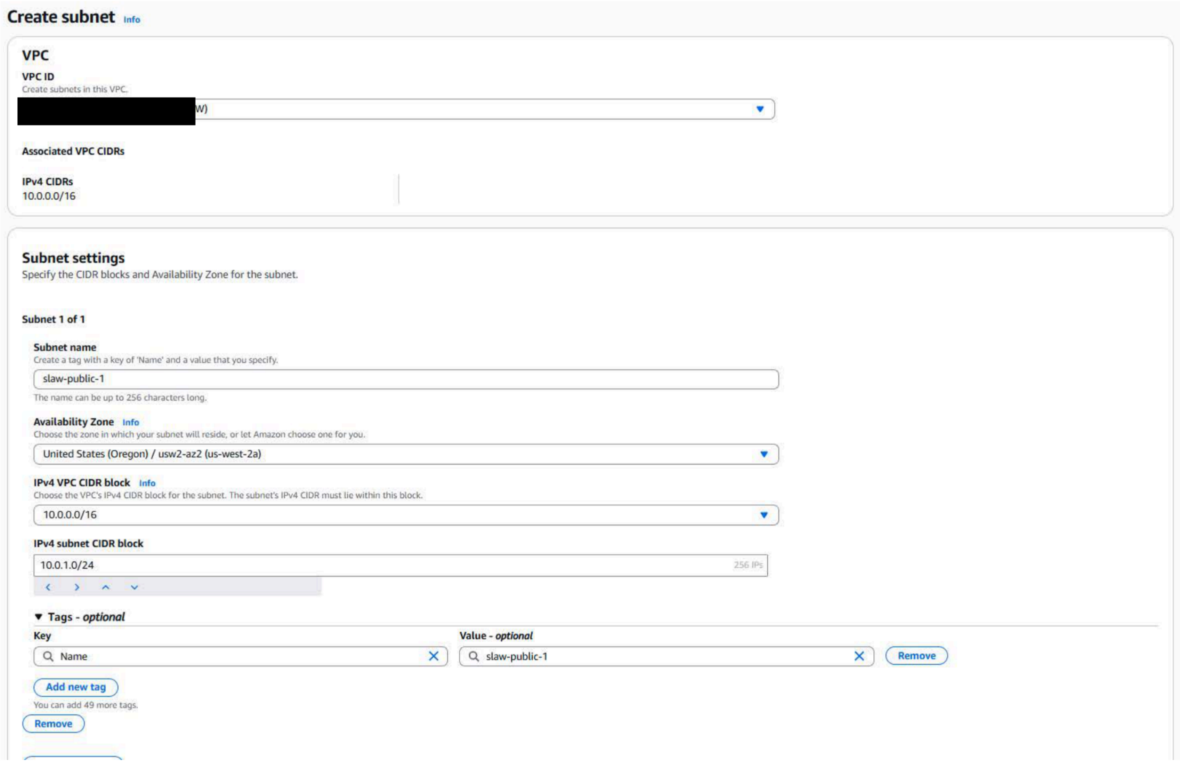
Evidence

Figure 2 - CloudSLAW VPC Settings



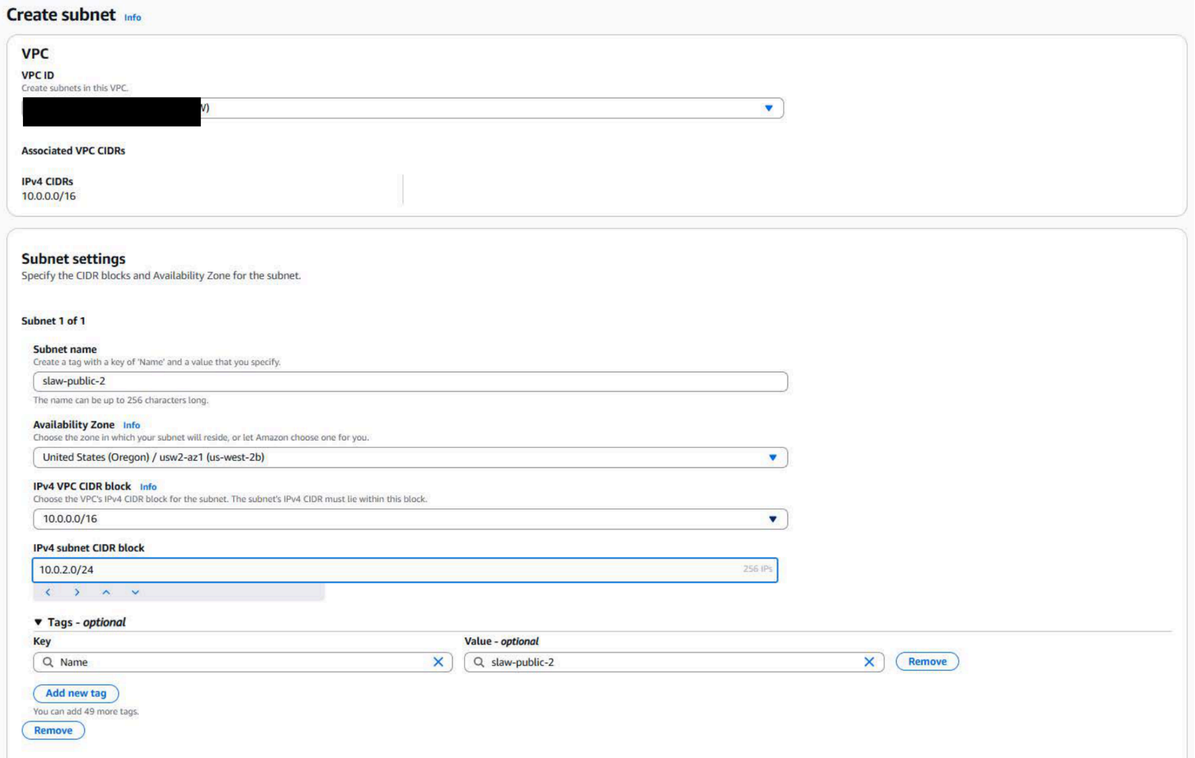
The VPC settings evidence shows VPC only selected, name tag CloudSLAW, manual IPv4 CIDR input set to 10.0.0.0/16, no IPv6 CIDR block, default tenancy, and VPC encryption control set to None.

Figure 3 - slaw-public-1 Subnet Configuration



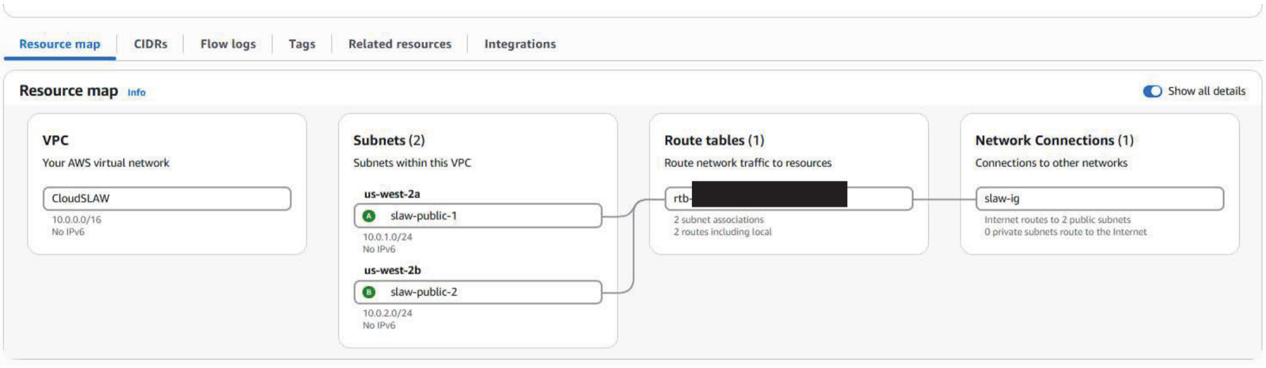
The subnet evidence shows the CloudSLAW VPC selected with associated IPv4 CIDR 10.0.0.0/16. Subnet slaw-public-1 is configured in us-west-2a with IPv4 subnet CIDR 10.0.1.0/24.

Figure 4 - slaw-public-2 Subnet Configuration



The subnet evidence shows the CloudSLAW VPC selected with associated IPv4 CIDR 10.0.0.0/16. Subnet slaw-public-2 is configured in us-west-2b with IPv4 subnet CIDR 10.0.2.0/24.

Figure 5 - Completed VPC Resource Map



The completed VPC evidence shows CloudSLAW with CIDR 10.0.0.0/16, two subnets named slaw-public-1 and slaw-public-2, one route table with two subnet associations and two routes including local, and one network connection named slaw-ig. The diagram also indicates Internet routes to two public subnets.

Security Analysis

This project creates a deliberate VPC instead of relying on a default VPC. That improves clarity and makes future security design easier to document and reason about.

The two subnets are public because the associated route table sends 0.0.0.0/0 to the Internet Gateway. Public routing does not automatically make an instance reachable; security groups, network ACLs, and public IP assignment still affect reachability.

Using separate Availability Zones gives the network a basic multi-AZ layout. That is useful groundwork for resilient workloads even though this lab does not deploy compute resources yet.

The VPC does not yet include private subnets, NAT, flow logs, endpoint routing, or segmentation. Those should be added in later production-style network projects.

Lessons Learned

CIDR notation defines the size and address space of both the VPC and the subnets inside it.

A VPC by itself is not enough for useful workload placement; subnets are required before resources can be launched into specific Availability Zones.

An Internet Gateway must be both created and attached to the VPC before it can be used as a route target.

A public subnet is defined primarily by routing: a route table association that sends Internet-bound traffic to an Internet Gateway.

Naming resources such as CloudSLAW, slaw-public-1, slaw-public-2, and slaw-ig makes the resource map and later troubleshooting easier.

Production Improvements

- Add private subnets in at least two Availability Zones.

- Create separate route tables for public and private subnets.
- Add NAT Gateway or egress-only patterns only when private workloads require outbound Internet access.
- Enable VPC Flow Logs for network visibility.
- Add consistent tags for owner, environment, cost center, data classification, and project.
- Use infrastructure as code such as CloudFormation or Terraform for repeatable VPC builds.
- Design network ACL and security group baselines before deploying workloads.

Skills Demonstrated

Cloud Networking	Routing	Security Design
Amazon VPC	Route tables	Public subnet design
CIDR planning	Internet Gateway	Multi-AZ subnet placement
Manual build workflow	Resource map validation	Public network risk analysis

Resume Bullet

RESUME	Built a custom AWS VPC with a 10.0.0.0/16 CIDR, two /24 public subnets across separate Availability Zones, an attached Internet Gateway, and route-table updates for Internet-bound traffic through slaw-ig.

STAR Interview Story

Situation: After deleting the default VPC, the lab environment needed a deliberately designed VPC to support future network and workload projects.

Task: Build a bare-bones custom VPC from scratch with subnets and Internet routing.

Action: I created the CloudSLAW VPC with 10.0.0.0/16, created two /24 subnets in separate Availability Zones, created and attached slaw-ig, and added the 0.0.0.0/0 route to the route table.

Result: The account now has a functioning custom public VPC foundation that can support future private subnet, NAT, flow log, and workload segmentation labs.

Version History

Version	Release Date	Description
v1.0.0	July 21, 2026	Initial documented implementation. Created a custom CloudSLAW VPC, configured two /24 public subnets, created and attached slaw-ig Internet Gateway, updated the route table for Internet-bound traffic, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Build a VPC from Scratch.
- User-provided CloudSLAW VPC settings screenshot.
- User-provided slaw-public-1 subnet screenshot.
- User-provided slaw-public-2 subnet screenshot.
- User-provided completed VPC resource map screenshot.
- Project 27 - Default VPC Exploration and Cleanup.